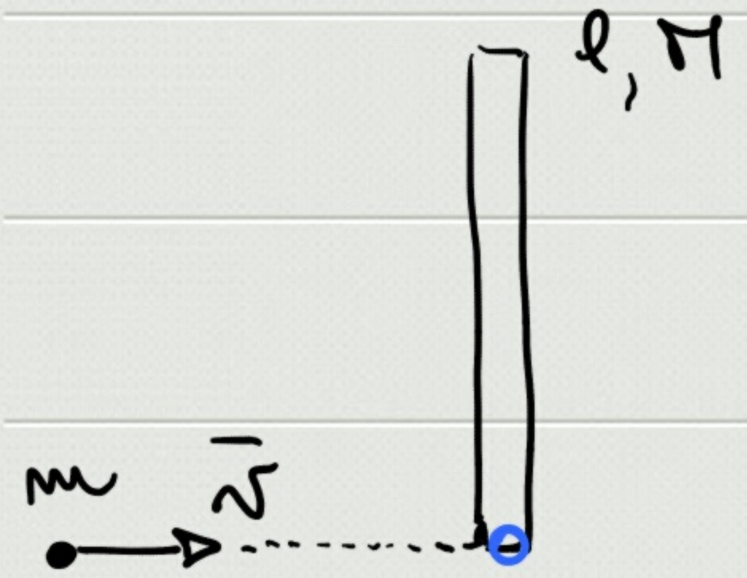


$v_\theta = \omega_0 = 0$

urto compl. anelastico

$\bar{v}', \omega'$

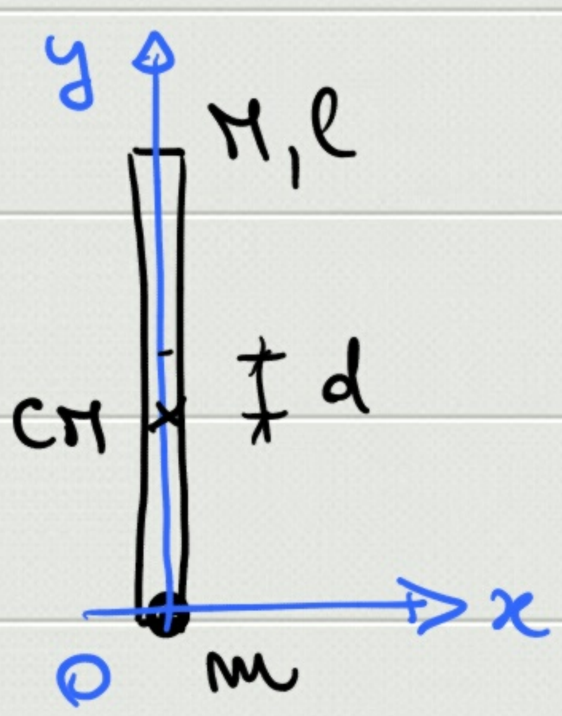
$\Uparrow$

$\boxed{\bar{P} = \text{cost}}$ $\boxed{\bar{L}^{\varepsilon}_{polo} = \text{cost}}$ $\bar{P} = m_T \bar{v}_{CM}$

$\bar{P}^- = \bar{P}^+$ $\bar{P}^- = m\bar{v}$

$\bar{P}^+ = (m+M)\bar{v}'_{CM}$

$$\Rightarrow \boxed{\bar{v}'_{CM} = \frac{m}{m+M}\bar{v}}$$



$$y_{CM} = \frac{\cancel{0}m + \frac{l}{2}M}{m+M} = \frac{M}{m+M}\frac{l}{2}$$

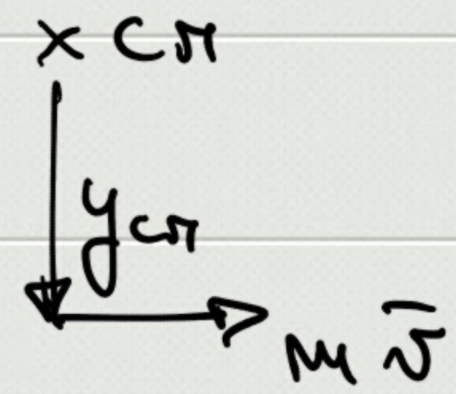
$$d = \frac{l}{2} - y_{CM} = \frac{m}{m+M}\frac{l}{2}$$

$\bar{L}^-_{CM} = \bar{L}^+_{CM}$

$$\bar{L} = \bar{r} \times m\bar{v} \qquad \bar{L} = I\bar{\omega}$$

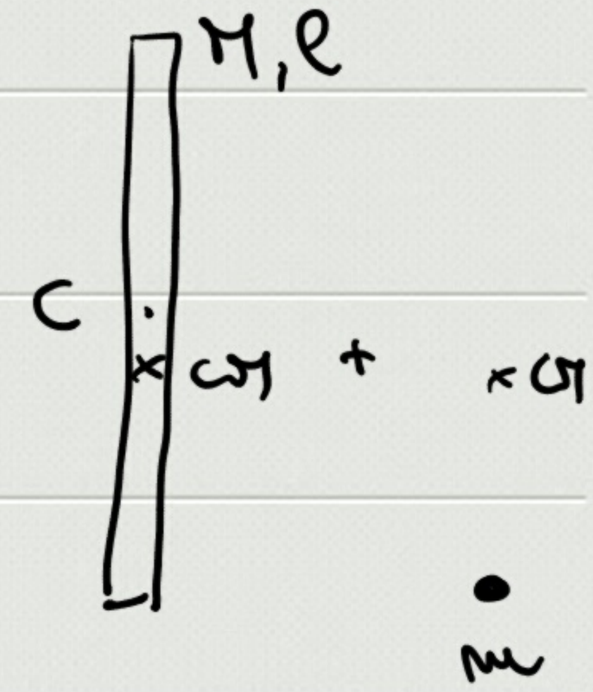
$$\bar{L}^{-}_{CM} = \bar{y}_{CM} \times m\bar{v}$$

$$L^{-}_{CM} = \frac{M}{m+M}\,\frac{\ell}{2}\,mv$$

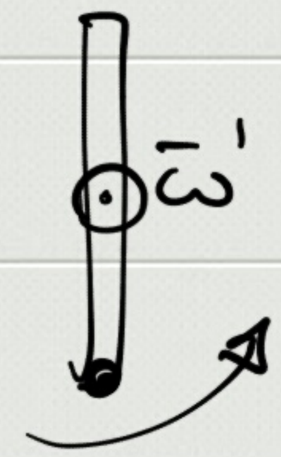


$$\bar{L}^{+}_{CM} = I_{CM}\,\bar{\omega}'$$

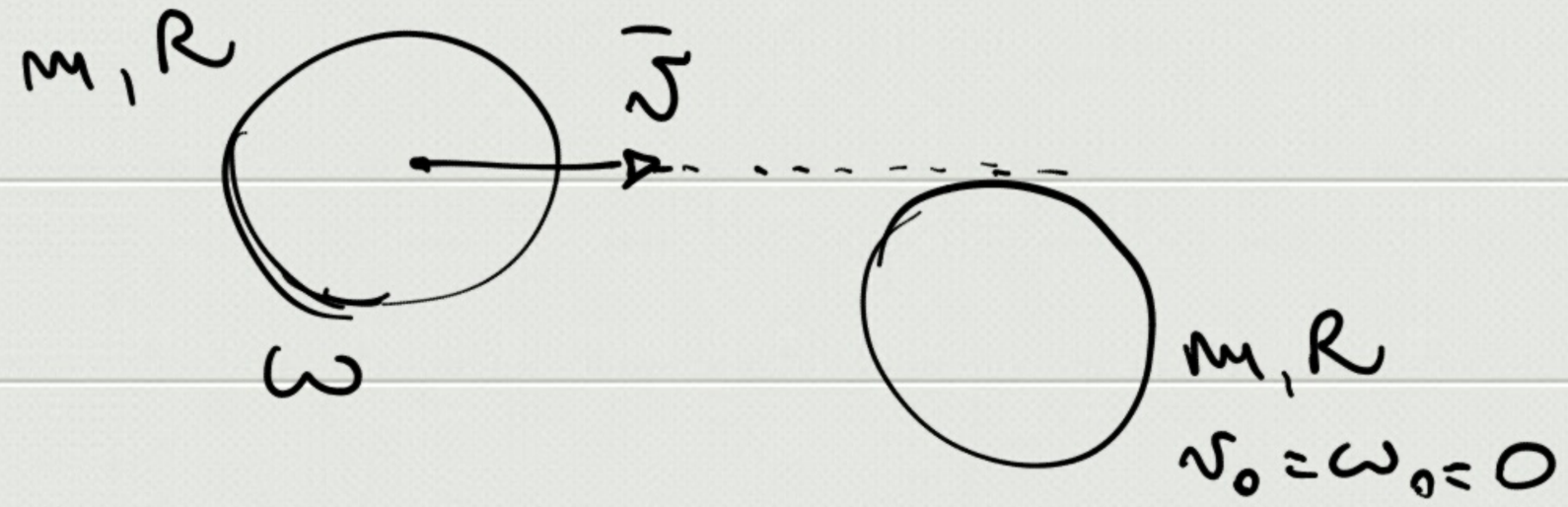
$$I_{CM} = \left(\frac{1}{12}M\ell^2 + Md^2\right) + m\,y^2_{CM}$$

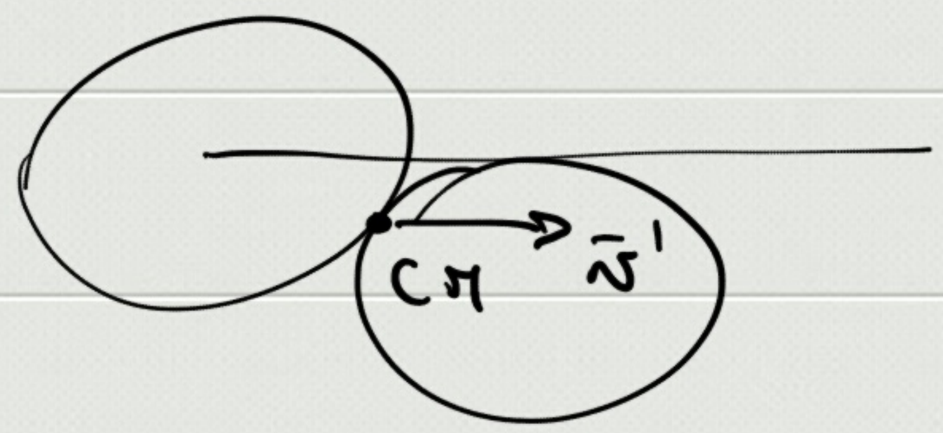


$$L^{-}_{CM} = L^{+}_{CM}$$



$$\Rightarrow \boxed{\omega' = \frac{6mv}{(4m+M)\ell}}$$





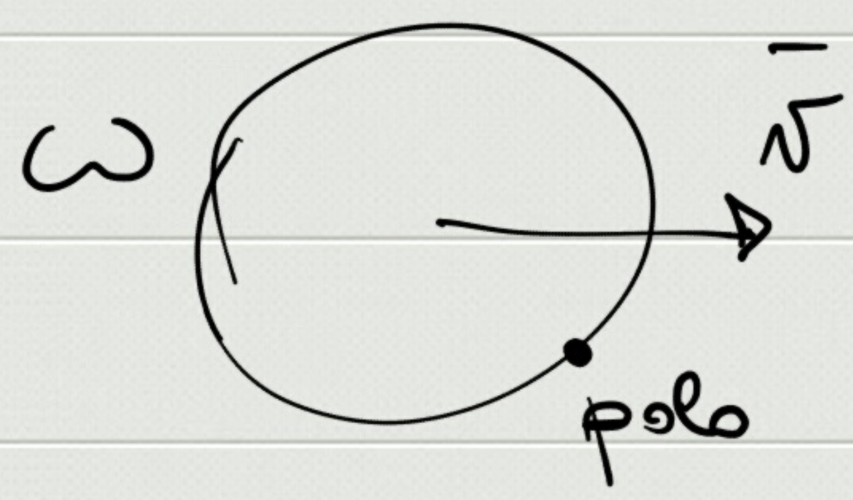
$$\boxed{\bar{v}' \qquad \bar{\omega}' = 0}$$

$$\bar{P} = \text{cost} \quad \Rightarrow \quad m\bar{v} = 2m\bar{v}'$$

$$\Rightarrow \boxed{\bar{v}' = \frac{\bar{v}}{2}}$$

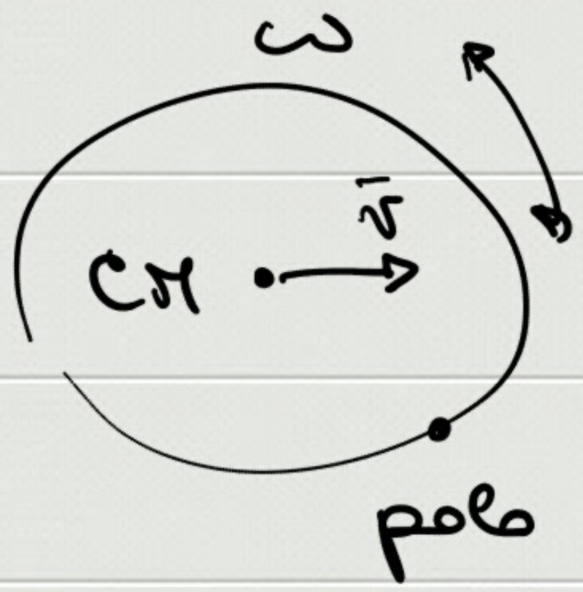
$$\bar{L}_{CM} = \text{cost} \qquad \bar{L}^{+}_{CM} = 0 \quad \Rightarrow \boxed{\bar{L}^{-}_{CM} = 0}$$

$$\bar{L}_0 = \bar{r} \times m\bar{v} \qquad \bar{L}_0 = \bar{I}_z \bar{\omega}$$

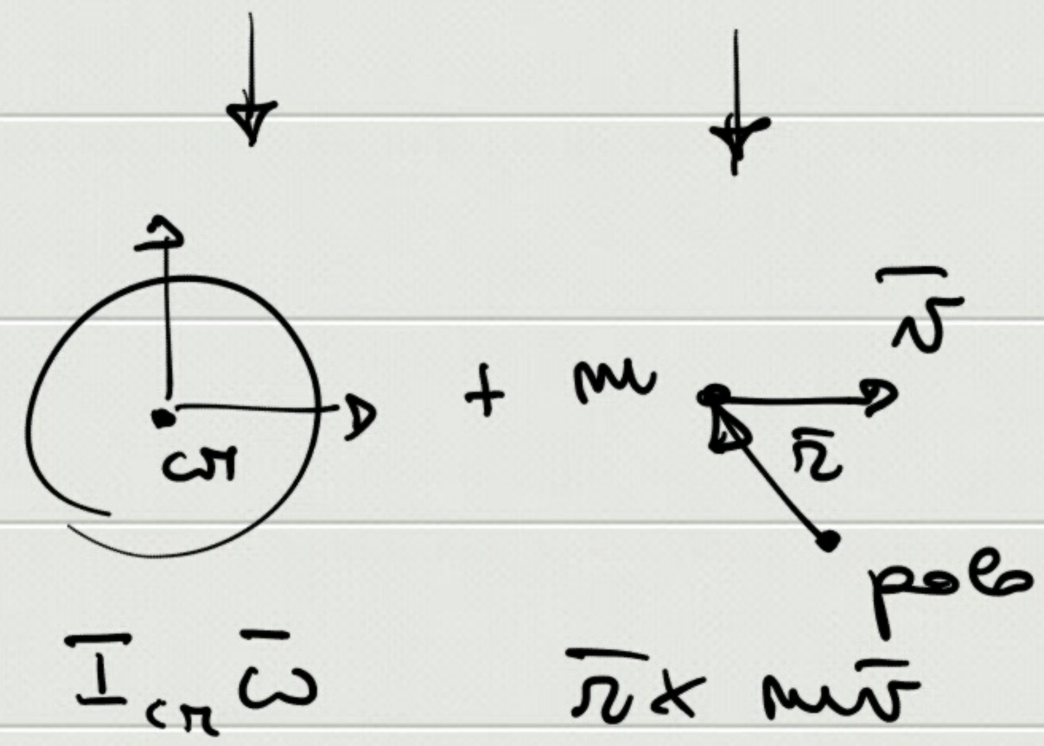


T. di Koenig per $\bar{L}$

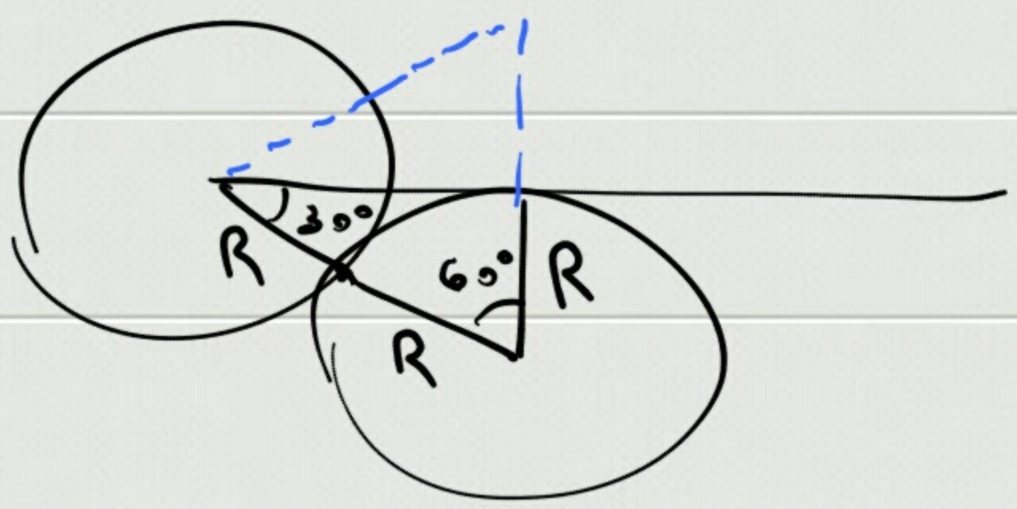
$$\boxed{\bar{L}_0 = \bar{L}'_{CM} + \bar{L}_{0,CM}}$$

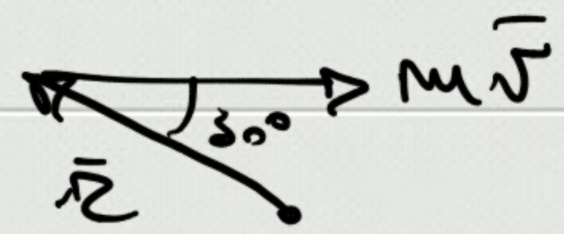


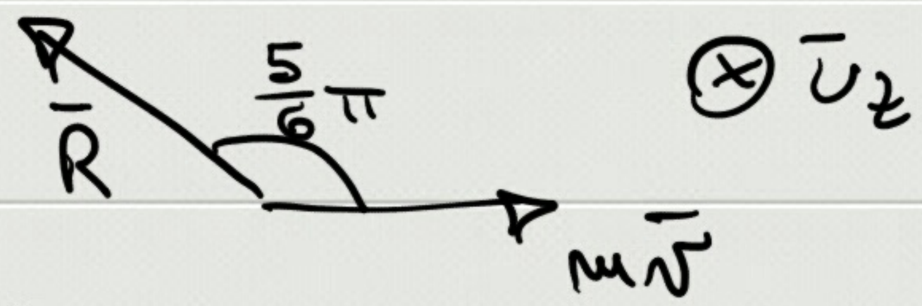
$$\bar{L}_{polo} = \bar{L}'_{CM} + \bar{L}_{polo,CM}$$



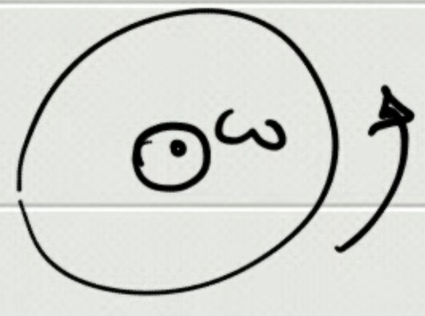
$$\Rightarrow \bar{L}_{polo} = \frac{1}{2} m R^2 \bar{\omega} + \bar{r} \times m\bar{v}$$







$$\bar{r} \times m\bar{v} = R m v \sin\frac{5}{6}\pi \bar{u}_z = \frac{1}{2} R m v \bar{u}_z$$

$$\Rightarrow \bar{L}_{polo} = \frac{1}{2} m R^2 \bar{\omega} + \frac{1}{2} R m v \bar{u}_z = 0$$

$$\Rightarrow \bar{\omega} = -\frac{v}{R}\bar{u}_z \qquad \boxed{v = \omega R}$$